

all-sites-exploratory

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Packages

```
library(tidyverse)
library(here)
library(janitor)
library(snakecase)
library(scales)
library(readxl)
library(patchwork)
library(ggribes)
library(lme4)
library(lmerTest)
library(knitr)
library(rmarkdown)
library(viridis)
library(viridisLite)
library(wesanderson)
library(ragg)
```

Data

```
# Trail Creek Isotope
compiled_isotope <- read_csv(here("data",
"compiled_river_reach_isotope_data.csv"), show_col_types = FALSE) |>
  clean_names() |> # lowercase and underscores in column names
  rename(sample = sample_name, # renaming columns to be shorter
    delta_d = processed_delta_d,
    delta_d_std = processed_delta_d_stddev,
    delta_o = processed_delta_18o,
    delta_o_std = processed_delta_18o_stddev) |>
  mutate(date = mdy(date)) # telling r the date format

gradient_data <- read_csv(here("data", "tc_gradient_data.csv"))

trail_creek <- compiled_isotope |>
```

```

filter(location %in% c("utc", "tc"))

coal_creek <- compiled_isotope |>
  filter(location == "cc")

italian_creek <- compiled_isotope |>
  filter(location == "ic")

```

Exploratory Visuals with ALL Sites

Line-Conditioned Excess (lc-excess) vs Delta 18O

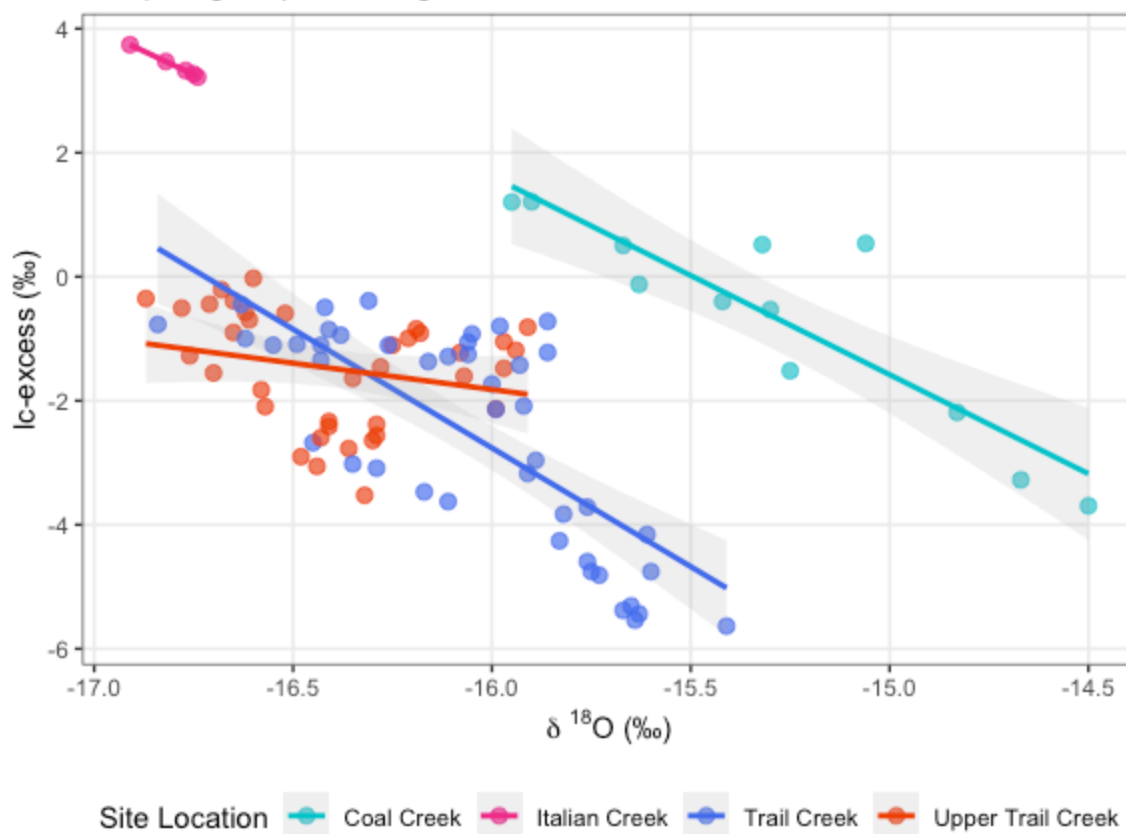
```

ggplot(data = compiled_isotope, aes(x = delta_o, y = lc_excess, color = location))
+
  geom_point(size = 2.5, alpha = 0.7) +
  geom_smooth(method = "lm", se = TRUE, alpha = 0.15, linewidth = 1) +
  scale_color_manual(
    values = c("tc" = "royalblue2", "cc" = "turquoise3", "ic" = "deeppink2",
"utc" = "orangered2"),
    labels = c("cc" = "Coal Creek", "ic" = "Italian Creek", "tc" = "Trail Creek",
"utc" = "Upper Trail Creek")
  ) +
  labs(
    title = "Line-Conditioned Excess (lc-excess) vs. Delta 180",
    subtitle = "Comparing evaporation signals between sites",
    x = expression(paste(delta, " ^18, "0 (\u2030)")), # Formats as  $\delta^{18}O$  (‰)
    y = "lc-excess (\u2030)",
    color = "Site Location") +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
    legend.position = "bottom",
    panel.grid.minor = element_blank())

```

Line-Conditioned Excess (lc-excess) vs. Delta 18O

Comparing evaporation signals between sites



Time-Series of Line-Conditioned Excess (lc-excess)

```
ggplot(data = compiled_isotope,
  aes(x = date,
    y = lc_excess,
    color = location,
    group = location)) +
  geom_hline(yintercept = 0,
    linetype = "dashed",
    color = "gray50",
    size = 0.8) +
  geom_point(size = 2.5,
    alpha = 0.8) +
  geom_line(linewidth = 1,
    alpha = 0.7) +
  labs(title = "Time-Series of Line-Conditioned Excess (lc-excess)",
    subtitle = "Values below 0 indicate potential evaporative enrichment",
```

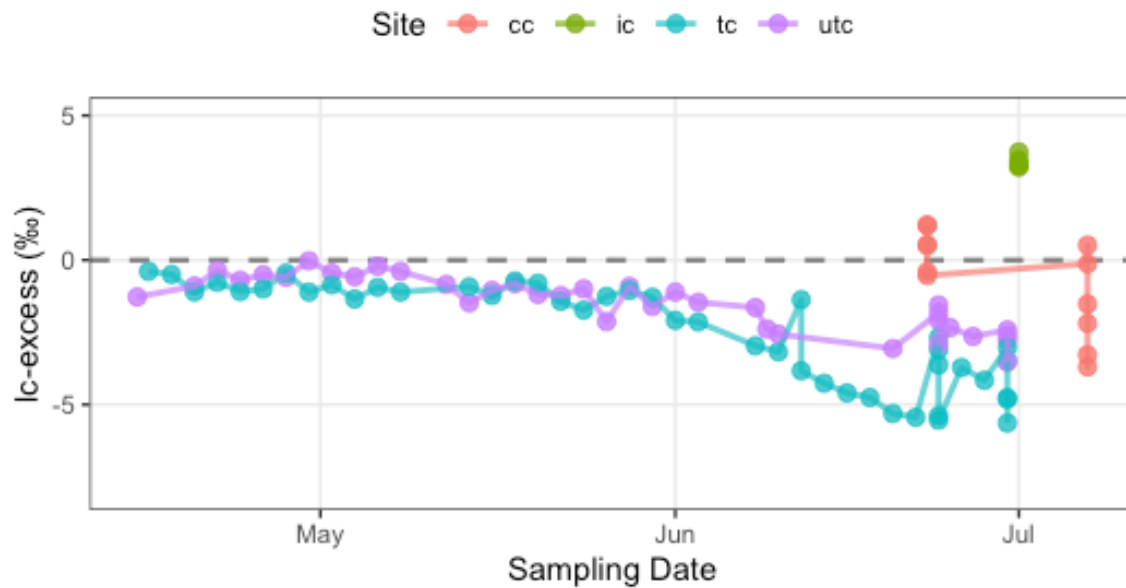
```

x = "Sampling Date",
y = "lc-excess (\u2030)",
color = "Site") +
ylim(-8,5) +
theme_bw() +
theme(plot.title = element_text(face = "bold", size = 14),
      panel.grid.minor = element_blank(),
      legend.position = "top")

```

Time-Series of Line-Conditioned Excess (lc-excess)

Values below 0 indicate potential evaporative enrichment



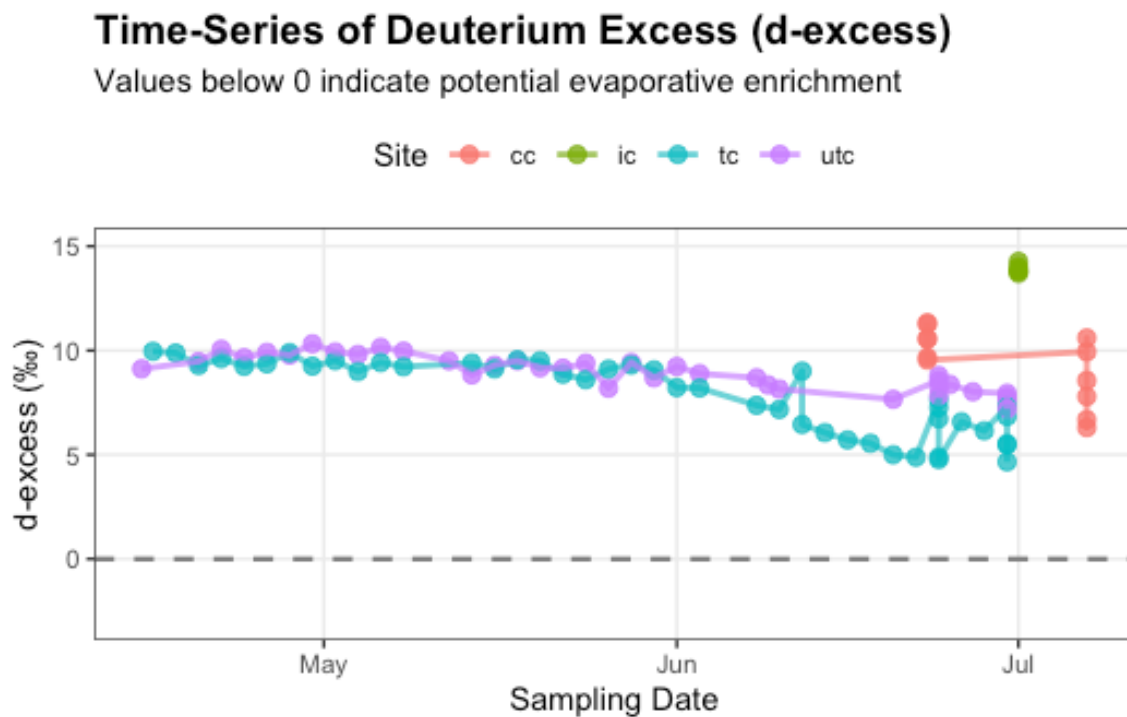
Time-Series of Deuterium Excess (d-excess)

```

ggplot(data = compiled_isotope,
       aes(x = date,
           y = d_excess,
           color = location,
           group = location)) +
geom_hline(yintercept = 0,
           linetype = "dashed",
           color = "gray50",
           size = 0.8) +
geom_point(size = 2.5,
           alpha = 0.8) +

```

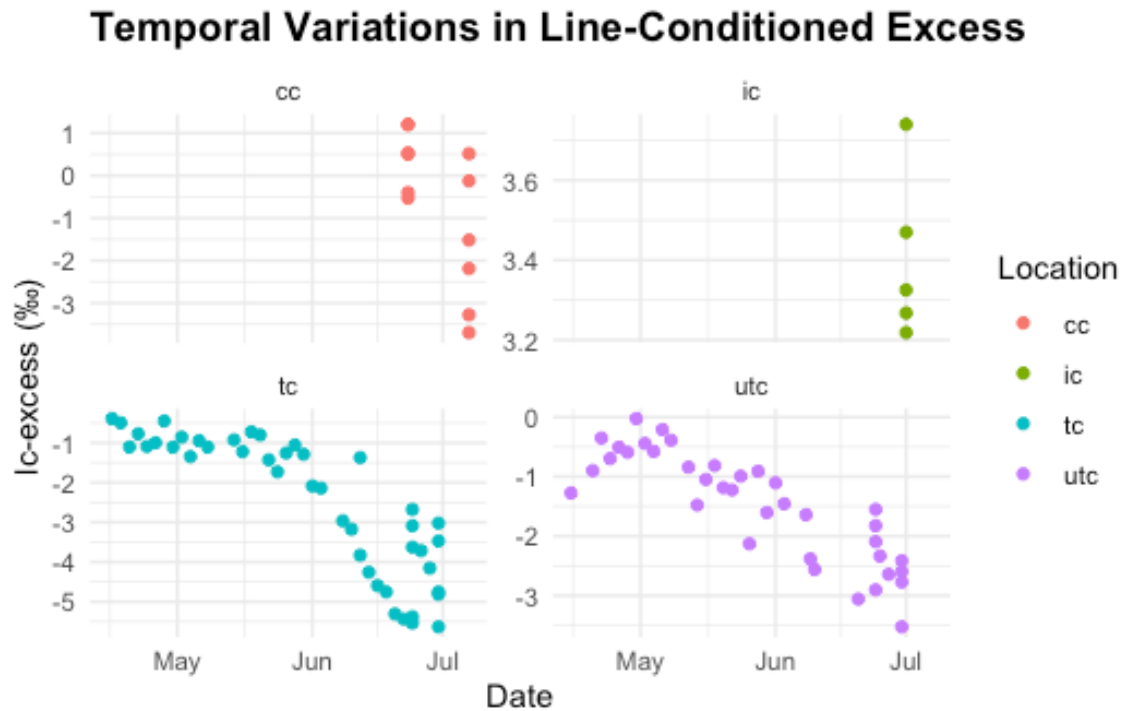
```
geom_line(size = 1,
          alpha = 0.7) +
labs(title = "Time-Series of Deuterium Excess (d-excess)",
     subtitle = "Values below 0 indicate potential evaporative enrichment",
     x = "Sampling Date",
     y = "d-excess (‰)",
     color = "Site") +
ylim(-3,15) +
theme_bw() +
theme(plot.title = element_text(face = "bold", size = 14),
      panel.grid.minor = element_blank(),
      legend.position = "top")
```



Temporal Variations in Line-Conditioned Excess (by location)

```
ggplot(compiled_isotope, aes(x = date, y = lc_excess, color = location)) +
geom_point(size = 1.5) +
labs(
  title = "Temporal Variations in Line-Conditioned Excess",
  x = "Date",
  y = "lc-excess (‰)",
  color = "Location"
```

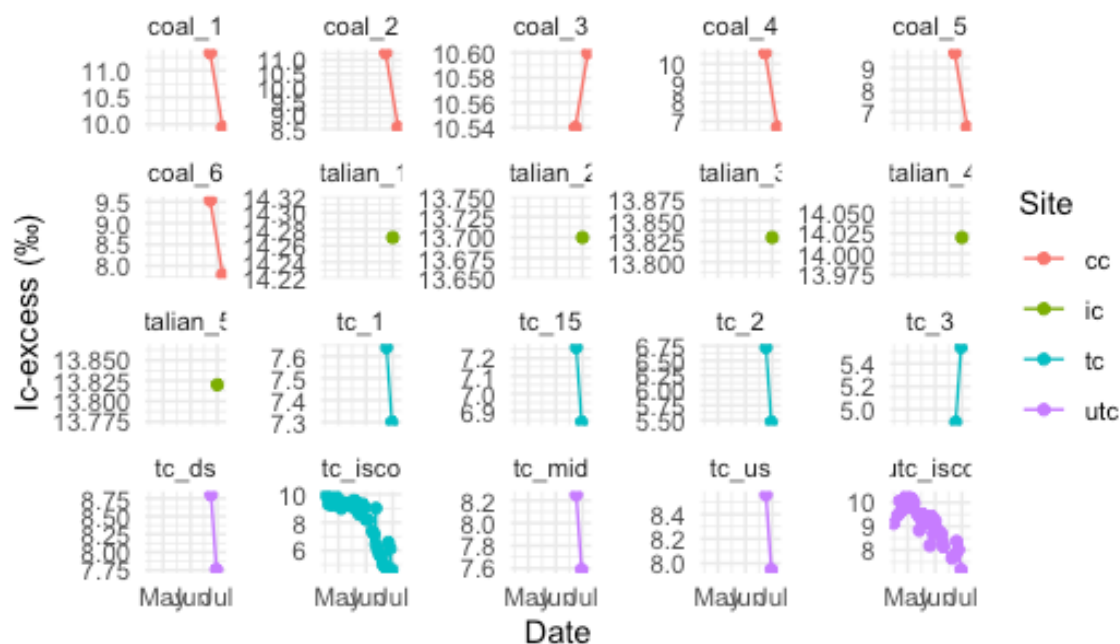
```
) +
facet_wrap(~location, scales = "free_y") + # Compare across sites easily
theme_minimal() +
theme(plot.title = element_text(face = "bold", size = 14))
```



Temporal Variations in Deuterium Excess (by sites)

```
ggplot(compiled_isotope, aes(x = date, y = d_excess, color = location)) +
  geom_line(linewidth = 0.5) +
  geom_point(size = 1.5) +
  labs(
    title = "Temporal Variations in Deuterium Excess",
    x = "Date",
    y = "lc-excess (‰)",
    color = "Site"
  ) +
  facet_wrap(~site, scales = "free_y") + # Compare across sites easily
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14))
```

Temporal Variations in Deuterium Excess

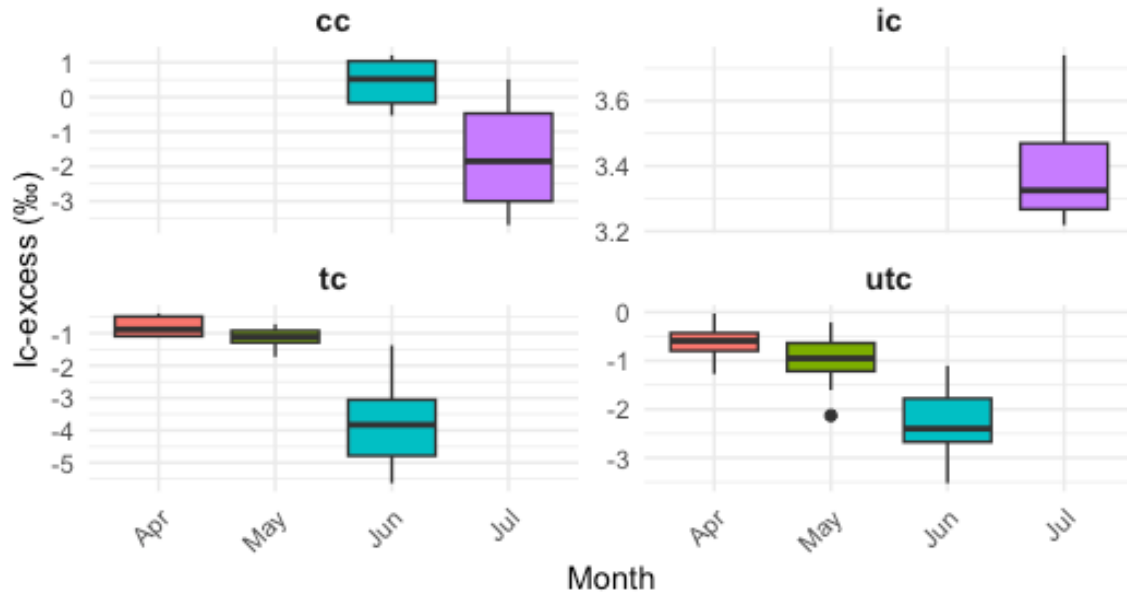


Monthly Variations in Evaporative Fraction (lc-excess) Across Study Sites

```
ggplot(compiled_isotope, aes(x = factor(month(date, label = TRUE, abbr = TRUE)),
y = lc_excess)) +
  geom_boxplot(aes(fill = factor(month(date))), show.legend = FALSE) +
  facet_wrap(~location, scales = "free_y") +
  labs(
    title = "Monthly Variations in Evaporative Fraction (lc-excess) Across
Study Sites",
    subtitle = "Seasonal isotopic shifts indicating changes in local evaporation-
condensation histories",
    x = "Month",
    y = "lc-excess (%)"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14),
    strip.text = element_text(face = "bold", size = 11),
    axis.text.x = element_text(angle = 45, hjust = 1)) # tilting the x axis text
```

Monthly Variations in Evaporative Fraction (lc-excess)

Seasonal isotopic shifts indicating changes in local evaporation-condensation



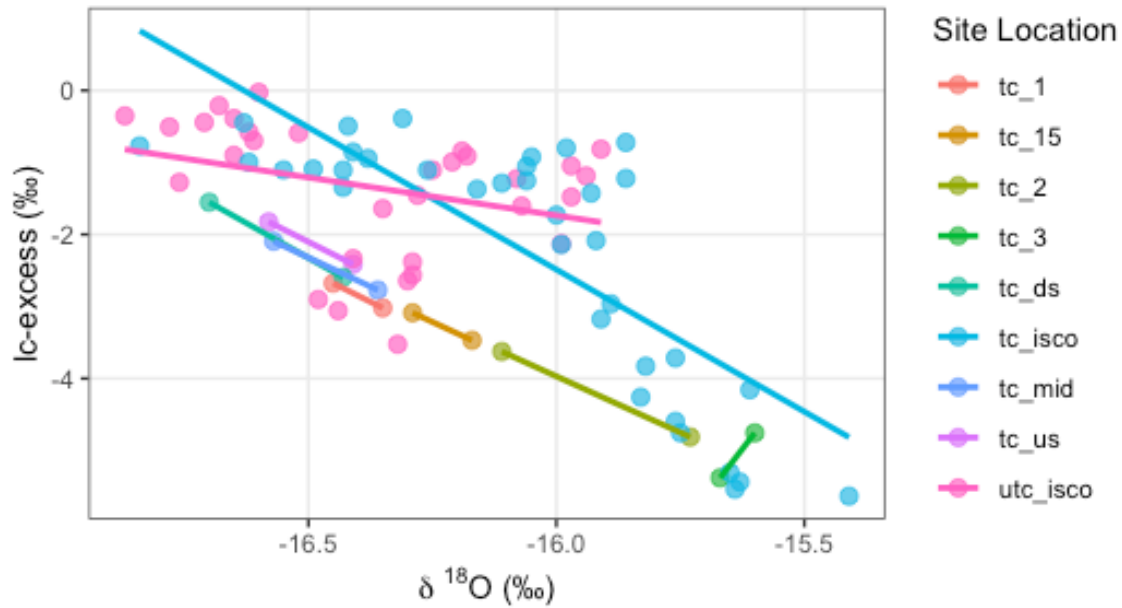
Exploratory Visuals of Trail Creek

Line-Conditioned Excess (lc-excess) vs. Delta 18O

```
ggplot(data = trail_creek, aes(x = delta_o, y = lc_excess, color = site)) +
  geom_point(size = 2.5, alpha = 0.7) +
  geom_smooth(method = "lm", se = FALSE, alpha = 0.15, linewidth = 1) +
  labs(
    title = "Line-Conditioned Excess (lc-excess) vs. Delta 180",
    subtitle = "Comparing evaporation signals between sites",
    x = expression(paste(delta, " ^18, " "0 (\u2030)")), # Formats as  $\delta^{18}O$  (‰)
    y = "lc-excess (\u2030)",
    color = "Site Location") +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        legend.position = "right",
        panel.grid.minor = element_blank())
```

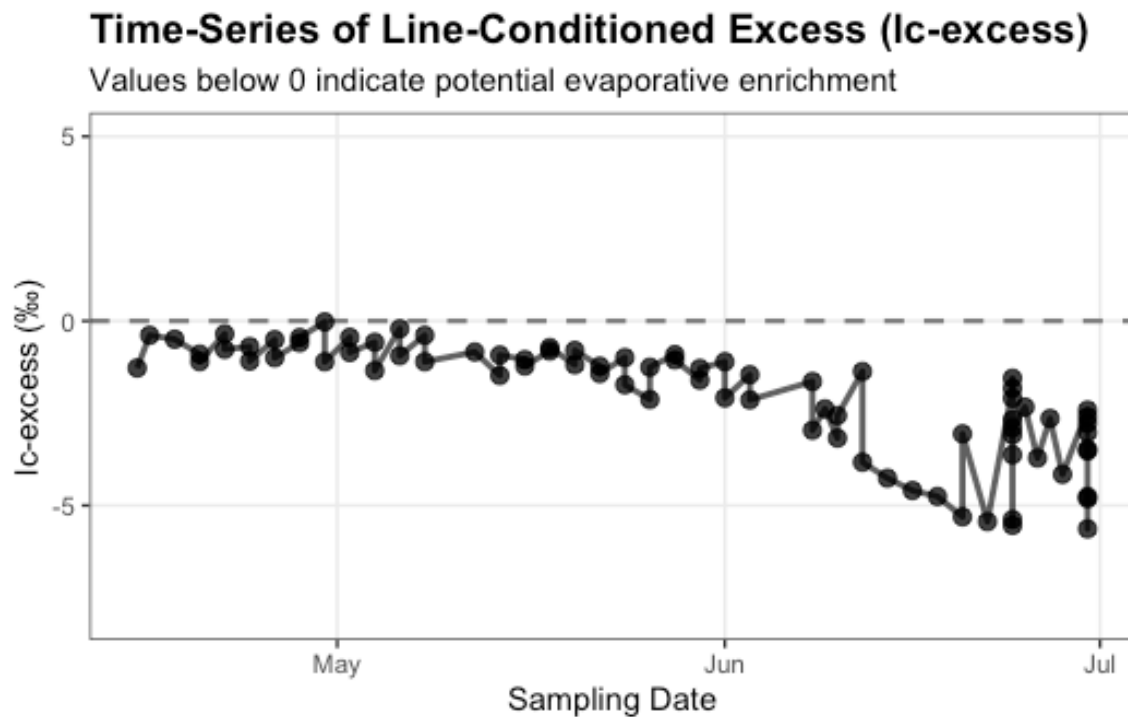

Line-Conditioned Excess (lc-excess) vs. Delta 18O

Comparing evaporation signals between sites



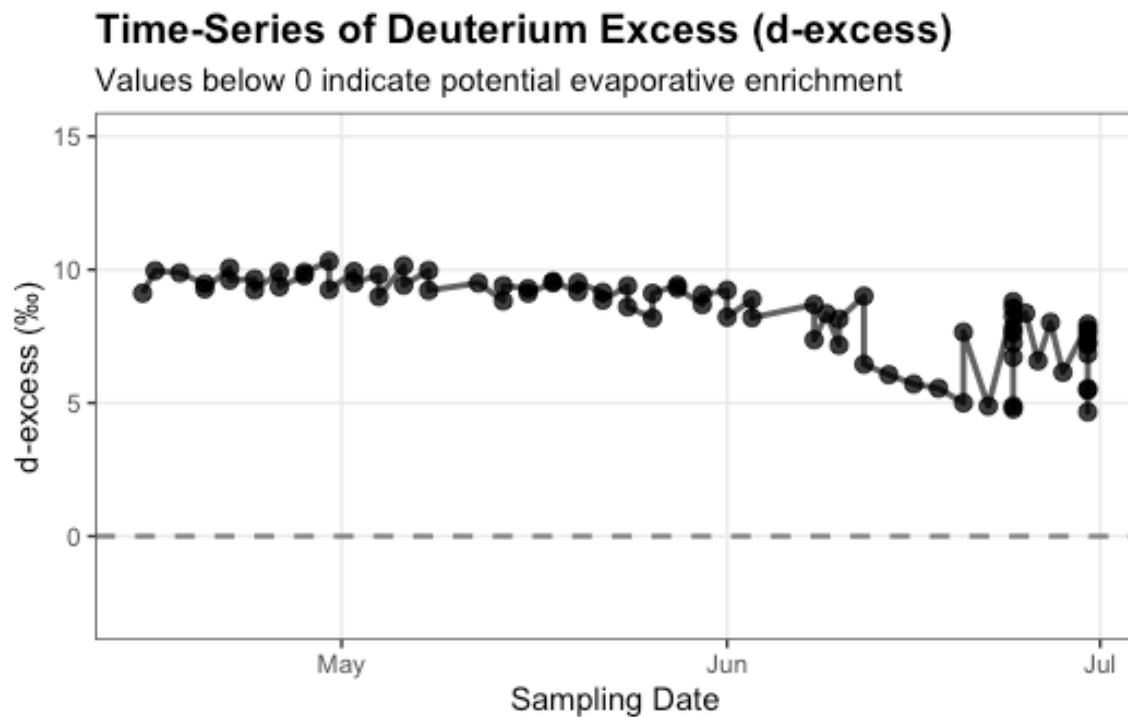
Time-Series of Line-Conditioned Excess (lc-excess)

```
ggplot(data = trail_creek,
       aes(x = date,
           y = lc_excess)) +
  geom_hline(yintercept = 0,
            linetype = "dashed",
            color = "gray50",
            size = 0.8) +
  geom_point(size = 2.5,
            alpha = 0.8) +
  geom_line(size = 1,
            alpha = 0.7) +
  labs(title = "Time-Series of Line-Conditioned Excess (lc-excess)",
       subtitle = "Values below 0 indicate potential evaporative enrichment",
       x = "Sampling Date",
       y = "lc-excess (‰)",
       color = "Site") +
  ylim(-8,5) +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        panel.grid.minor = element_blank(),
        legend.position = "top")
```



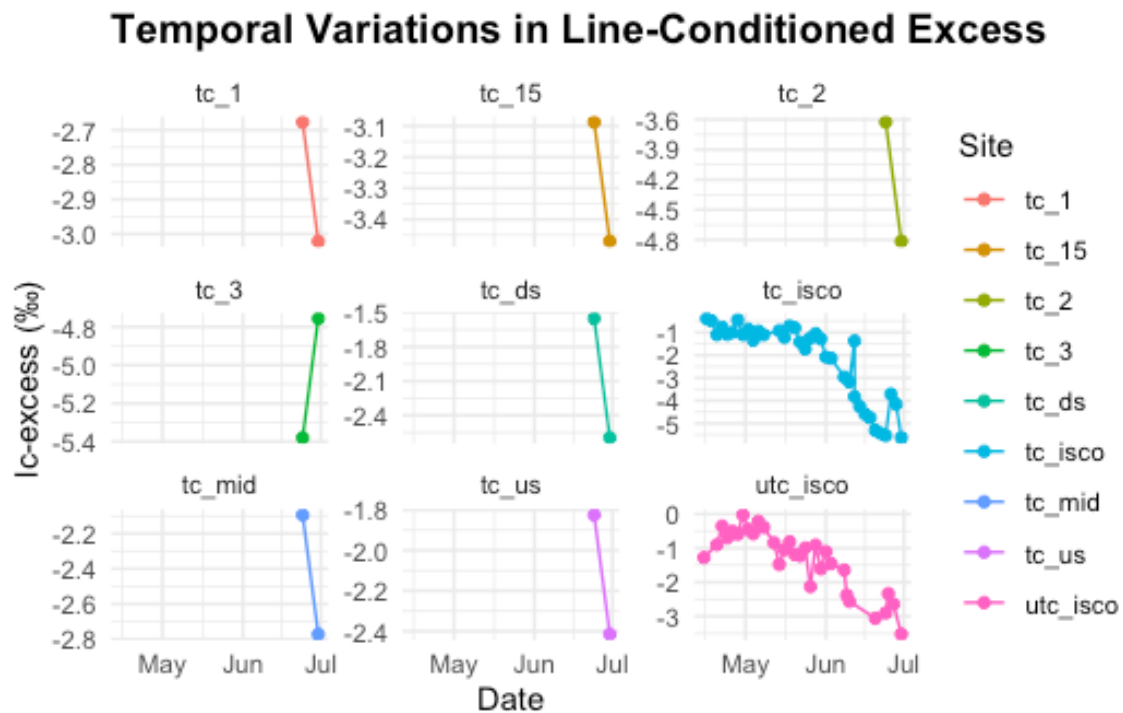
Time-Series of Deuterium Excess (d-excess)

```
ggplot(data = trail_creek,
  aes(x = date,
    y = d_excess)) +
  geom_hline(yintercept = 0,
    linetype = "dashed",
    color = "gray50",
    size = 0.8) +
  geom_point(size = 2.5,
    alpha = 0.8) +
  geom_line(size = 1,
    alpha = 0.7) +
  labs(title = "Time-Series of Deuterium Excess (d-excess)",
    subtitle = "Values below 0 indicate potential evaporative enrichment",
    x = "Sampling Date",
    y = "d-excess (‰)",
    color = "Site") +
  ylim(-3,15) +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
    panel.grid.minor = element_blank(),
    legend.position = "top")
```



Temporal Variations in Line-Conditioned Excess

```
ggplot(trail_creek, aes(x = date, y = lc_excess, color = site)) +
  geom_line(linewidth = 0.5) +
  geom_point(size = 1.5) +
  labs(
    title = "Temporal Variations in Line-Conditioned Excess",
    x = "Date",
    y = "lc-excess (%)",
    color = "Site"
  ) +
  facet_wrap(~site, scales = "free_y") + # Compare across sites easily
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14))
```

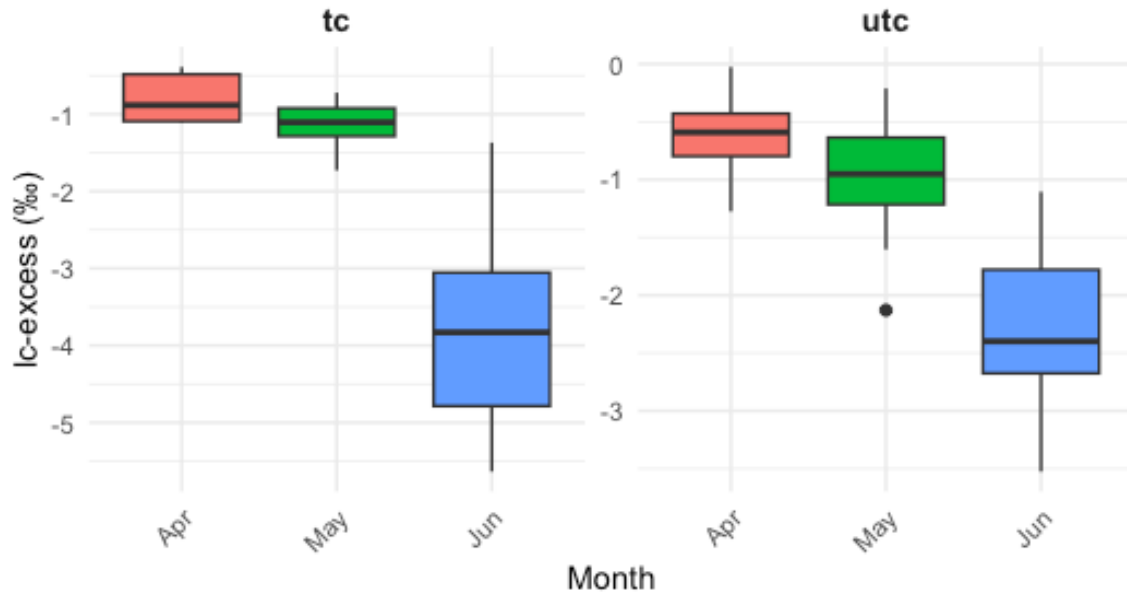


Monthly Variations in Evaporative Fraction (lc-excess) Across Study Sites

```
ggplot(trail_creek, aes(x = factor(month(date, label = TRUE, abbr = TRUE)), y =
lc_excess)) +
  geom_boxplot(aes(fill = factor(month(date))), show.legend = FALSE) +
  facet_wrap(~location, scales = "free_y") +
  labs(
    title = "Monthly Variations in Evaporative Fraction (lc-excess) Across
Study Sites",
    subtitle = "Seasonal isotopic shifts indicating changes in local evaporation-
condensation histories",
    x = "Month",
    y = "lc-excess (‰)"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14),
    strip.text = element_text(face = "bold", size = 11), # Emphasize site names
    axis.text.x = element_text(angle = 45, hjust = 1) # Tilt month labels if
they overlap
  )
```

Monthly Variations in Evaporative Fraction (lc-excess)

Seasonal isotopic shifts indicating changes in local evaporation-condensation



Dual-isotope plot ($\delta^{18}\text{O}$ vs δD) with LMWL reference

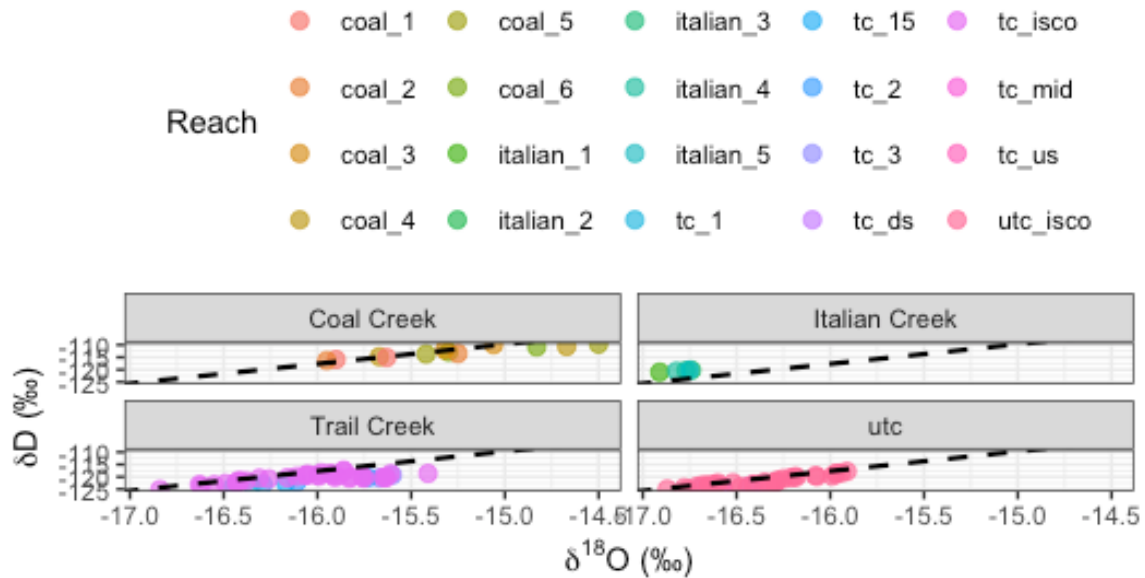
```
# Trail Creek LMWL calculated from OPIC

lmwl_slope <- 7.94
lmwl_intercept <- 9.37

ggplot(compiled_isotope, aes(x = delta_o, y = delta_d, color = site)) +
  geom_point(size = 2.5, alpha = 0.7) +
  geom_abline(slope = lmwl_slope, intercept = lmwl_intercept,
              linetype = "dashed", color = "black", linewidth = 0.8) +
  facet_wrap(~location, labeller = labeller(location = c(tc = "Trail Creek",
                                                         cc = "Coal Creek",
                                                         ic = "Italian
Creek")))) +
  labs(title = "Dual Isotope Plot:  $\delta^{18}\text{O}$  vs  $\delta\text{D}$ ",
       subtitle = "Dashed line = Trail Creek LMWL",
       x = expression(paste(delta^18, "O (‰)")),
       y = expression(paste(delta, "D (‰)")),
       color = "Reach") +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        legend.position = "top")
```

Dual Isotope Plot: $\delta^{18}\text{O}$ vs δD

Dashed line = Trail Creek LMWL



Time series of lc-excess across season, colored by reach, faceted by watershed

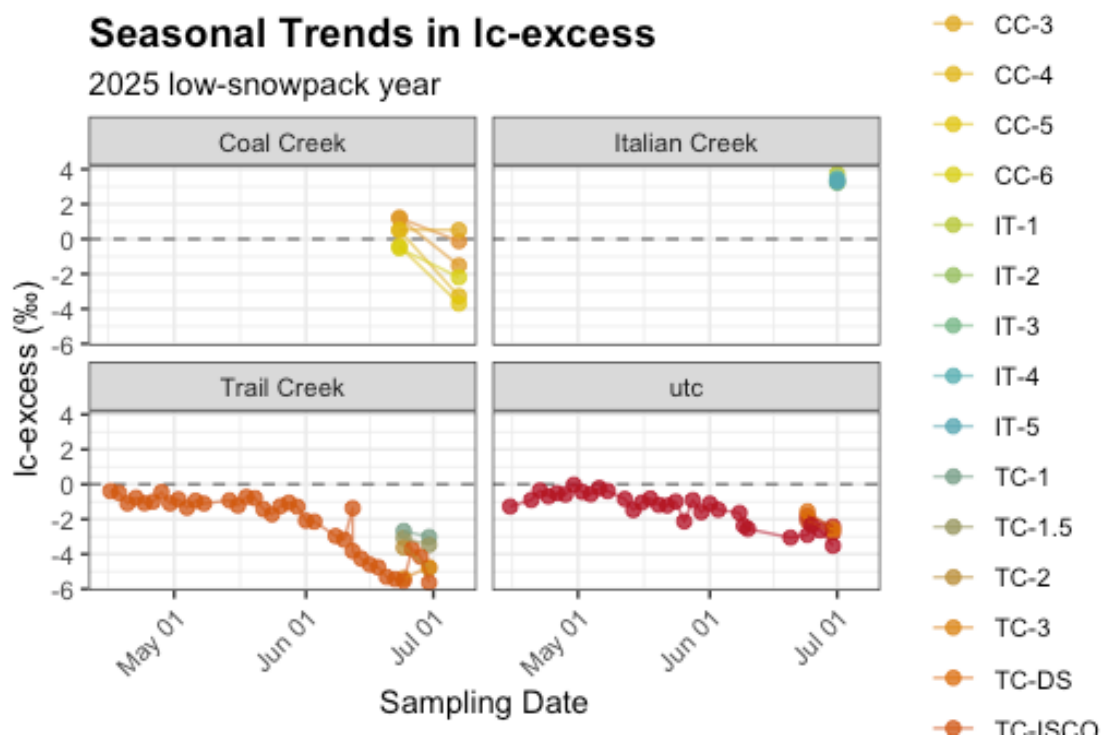
```
ggplot(compiled_isotope, aes(x = date, y = lc_excess, color = site, group =
interaction(site, location))) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray50") +
  geom_point(size = 2, alpha = 0.8) +
  geom_line(alpha = 0.6) +
  facet_wrap(~location, labeller = labeller(location = c(tc = "Trail Creek",
                                                         cc = "Coal Creek",
                                                         ic = "Italian Creek")))) +

  scale_x_date(date_labels = "%b %d") +
  scale_color_manual(values = wes_palette("FantasticFox1", n =
length(unique(compiled_isotope$site)),
                    type = "continuous"),
                    labels = c("coal_1" = "CC-1", "coal_2" = "CC-2", "coal_3"
= "CC-3", "coal_4" =
"CC-4", "coal_5" = "CC-5", "coal_6" = "CC-6",
"italian_1" = "IT-1",
"italian_2" = "IT-2", "italian_3" = "IT-3",
"italian_4" = "IT-4",
"italian_5" = "IT-5", "tc_isco" = "TC-ISCO",
"tc_1" = "TC-1", "tc_15"
= "TC-1.5", "tc_2" = "TC-2", "tc_3" = "TC-3",
```

```

"tc_ds" = "TC-DS",
                                "tc_mid" = "TC-Mid", "tc_us" = "TC-US", "utc_isco"
= "UTC-ISCO")) +
  labs(title = "Seasonal Trends in lc-excess",
        subtitle = "2025 low-snowpack year",
        x = "Sampling Date", y = "lc-excess (‰)", color = "Reach") +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        legend.position = "right",
        axis.text.x = element_text(angle = 45, hjust = 1))

```



```

# averaging the lc-excess and including standard error
delta_summary <- compiled_isotope |>
  group_by(site) |>
  summarize(mean_lc = mean(lc_excess, na.rm = TRUE),
            se_lc = sd(lc_excess, na.rm = TRUE) / sqrt(n()),
            .groups = "drop")

# creating a plot to show the mean lc-excess of each site
ggplot(delta_summary, aes(x = site, y = mean_lc, fill = site)) +
  geom_col(color = "black", width = 0.7) +
  geom_errorbar(aes(ymin = mean_lc - se_lc, ymax = mean_lc + se_lc),

```

```

        width = 0.3,
        color = "black") +
    scale_x_discrete(limits = c("italian_1", "italian_2", "italian_3",
    "italian_4", "italian_5", "coal_1", "coal_2", "coal_3", "coal_4", "coal_5",
    "coal_6", "tc_isco", "tc_1", "tc_15", "tc_2", "tc_3", "utc_isco", "tc_ds",
    "tc_mid", "tc_us"),
    labels = c("coal_1" = "CC-1", "coal_2" = "CC-2", "coal_3" = "CC-3", "coal_4"
    = "CC-4",
        "coal_5" = "CC-5", "coal_6" = "CC-6", "italian_1" = "IT-1",
    "italian_2" = "IT-2",
        "italian_3" = "IT-3", "italian_4" = "IT-4", "italian_5" = "IT-5",
    "tc_isco" = "TC-ISCO", "tc_1" = "TC-1", "tc_15" = "TC-1.5", "tc_2"
    = "TC-2",
        "tc_3" = "IT-3", "tc_ds" = "IT-DS", "tc_mid" = "TC-Mid", "tc_us" =
    "TC-US",
        "utc_isco" = "UTC-ISCO")) +
    scale_fill_viridis_d() + # color palette
    theme_minimal() +
    labs(title = "Mean LC Excess by Site with Standard Error",
        x = "Sampling Site",
        y = "Mean LC Excess") +
    theme(plot.title = element_text(face = "bold", size = 14),
        axis.text.x = element_text(angle = 45, hjust = 1), # tilt labels
        legend.position = "none")

```




```
# Building a look up table of stream distances
distances <- tibble(site = c("TC-P3-US", "TC-P3-mid", "TC-P3-DS", "TC-BDA1",
                             "TC-BDA1.5", "TC-BDA2", "TC-BDA3"),
                    dist_m = c(57.6, 333, 654, 1160, 1620, 2240, 3150)) |>
mutate(xmax = dist_m,
       xmin = lag(dist_m, default = 0)) # previous site's distance, 0 for first

# Creating a filtered data frame for lc-excess
gradient_lc_clean <- gradient_data |>
  select(site, normalized_stream_distance_m, `Δlc-excess/Δx_6-24`, `Δlc-excess/
Δx_6-30`) |>
  pivot_longer( cols = c(`Δlc-excess/Δx_6-24`, `Δlc-excess/Δx_6-30`),
                names_to = "date",
                values_to = "gradient") |>
  mutate(date = case_when(date == "Δlc-excess/Δx_6-24" ~ "2026-06-24",
                          date == "Δlc-excess/Δx_6-30" ~ "2026-06-30"),
         gradient = as.numeric(gradient)) |>
  left_join(distances %>% select(site, xmin, xmax), by = "site") |>
  mutate(site = factor(site, levels = distances$site)) |>
  filter(!is.na(gradient))

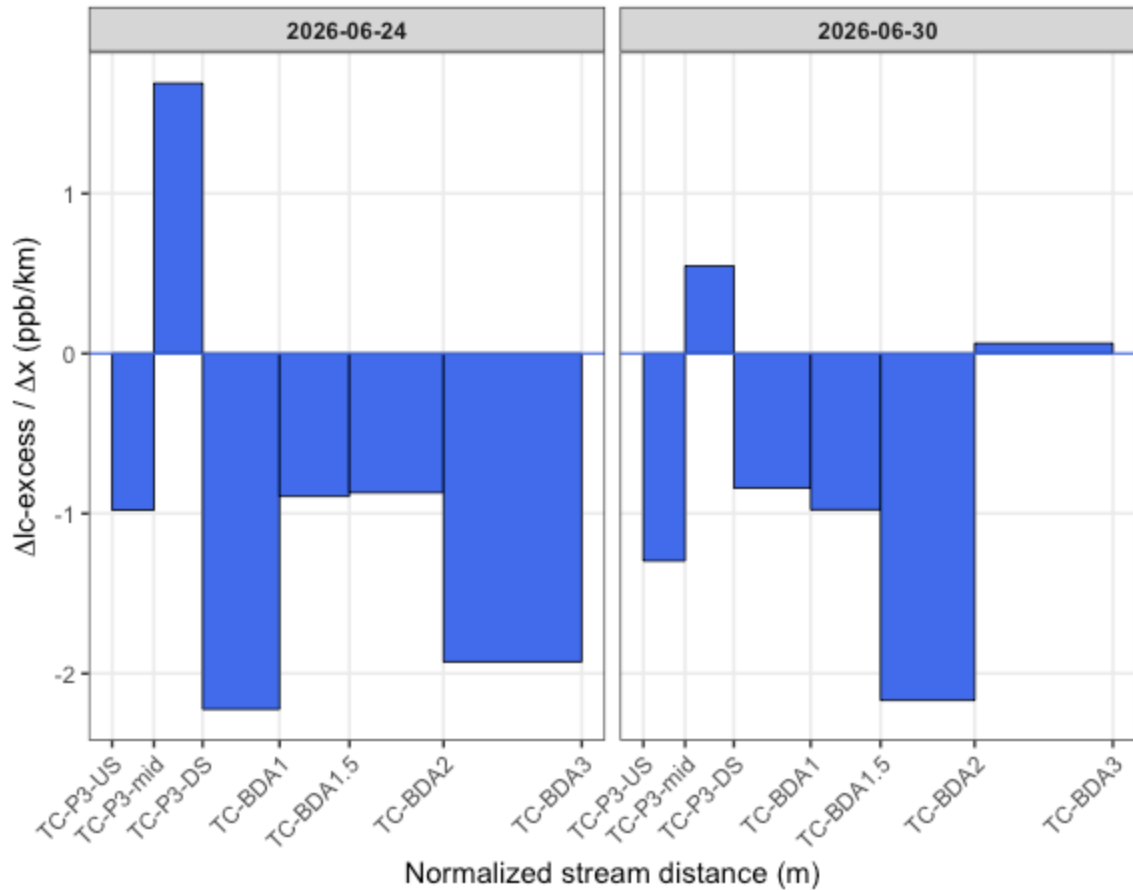
# Gradient plot
```

```

ggplot(gradient_lc_clean) +
  geom_rect(aes(xmin = xmin, xmax = xmax, ymin = 0, ymax = gradient),
            fill = "royalblue2", color = "black", linewidth = 0.3) +
  geom_hline(yintercept = 0, color = "royalblue2", linewidth = 0.4) +
  facet_wrap(~ date, nrow = 1) +
  scale_x_continuous(
    breaks = distances$dist_m,
    labels = distances$site) +
  labs(title = "Trail Creek: Spatial variations in  $\Delta$ lc-excess/ $\Delta$ x across sampling
dates",
       x = "Normalized stream distance (m)",
       y = expression(paste(Delta, "lc-excess / ", Delta, "x (ppb/km)"))) +
  theme_bw(base_size = 11) +
  theme(plot.title = element_text(face = "bold", size = 14),
        panel.grid.minor = element_blank(),
        strip.text = element_text(face = "bold"),
        axis.text.x = element_text(angle = 45, hjust = 1))

```

Trail Creek: Spatial variations in $\Delta lc\text{-excess}/\Delta x$ across samp



All reaches show lc-excess decreasing downstream, on both dates. Since lc-excess becoming more negative = more evaporative enrichment, this version says every single reach along Trail Creek is adding evaporative signal, with no recovery zones. That's a materially different story than "some reaches enrich, some reaches dilute" — it now reads as continuous, cumulative evaporative loss along the whole restored gradient.

Where the deepest enrichment is happening differs by date:

6-24: TC-P3-DS reach is the sharpest dip (~ -2.2), essentially a local hotspot just downstream of the mid-P3 restoration point. 6-30: TC-BDA1.5→BDA2 is now the sharpest dip (~ -2.15), while TC-P3-DS has relaxed to a much smaller magnitude (~ -0.55) compared to 6-24.

TC-BDA3 is the outlier reach on both dates — it's close to flat (near-zero gradient), meaning that final reach adds almost no additional evaporative signal, regardless of date. That's consistent between both panels and is probably your most robust finding here: whatever's happening upstream, the lowest reach isn't contributing much additional fractionation.

```

gradient_dams_clean <- gradient_data %>%
  select(site, cumulative_dams,
         `Δlc-excess/Δx_6-24`, `Δlc-excess/Δx_6-30`) %>%
  pivot_longer(
    cols = c(`Δlc-excess/Δx_6-24`, `Δlc-excess/Δx_6-30`),
    names_to = "date",
    values_to = "gradient"
  ) %>%
  mutate(
    date = case_when(
      date == "Δlc-excess/Δx_6-24" ~ "2026-06-24",
      date == "Δlc-excess/Δx_6-30" ~ "2026-06-30"
    ),
    gradient = as.numeric(gradient),
    cumulative_dams = as.numeric(cumulative_dams),
    site = factor(site, levels = c("TC-P3-US", "TC-P3-mid", "TC-P3-DS",
                                   "TC-BDA1", "TC-BDA1.5", "TC-BDA2", "TC-BDA3"))
  ) %>%
  filter(!is.na(gradient))

```

Warning: There were 2 warnings in `mutate()`.

The first warning was:

i In argument: `gradient = as.numeric(gradient)`.

Caused by warning:

! NAs introduced by coercion

i Run `dplyr::last\_dplyr\_warnings()` to see the 1 remaining warning.

```

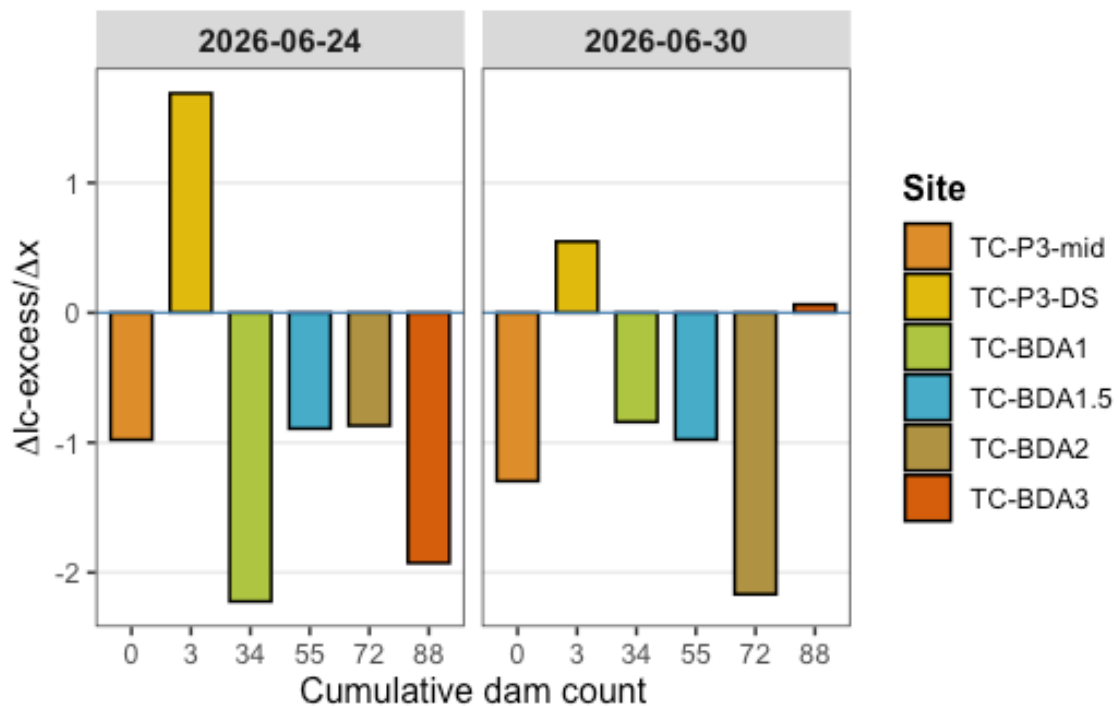
ggplot(gradient_dams_clean, aes(x = site, y = gradient, fill = site)) +
  geom_col(color = "black", width = 0.7) +
  geom_hline(yintercept = 0, color = "steelblue", linewidth = 0.4) +
  facet_wrap(~ date, nrow = 1) +
  scale_fill_manual(values = wes_palette("FantasticFox1", n = 7, type =
"continuous")) +
  scale_x_discrete(labels = function(x) {
    lookup <- setNames(gradient_dams_clean$cumulative_dams,
gradient_dams_clean$site)
    lookup[x]
  }) +
  labs(
    x = "Cumulative dam count",
    y = expression(Delta*"lc-excess/"*Delta*"x"),
    fill = "Site"
  )

```

```

) +
theme_bw(base_size = 12) +
theme(
  strip.text = element_text(face = "bold", size = 11),
  strip.background = element_rect(fill = "gray85", color = NA),
  panel.grid.minor = element_blank(),
  panel.grid.major.x = element_blank(),
  legend.title = element_text(face = "bold")
)

```



Exploratory Visuals of Coal Creek

Exploratory Visuals of Italian Creek